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Improving Independence: An Arduino Nano-Based Smart Stick for the Visually Impaired

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Abstract— **The Smart Blind Stick is a wearable device that helps visually impaired people move safely and independently. It is built on an Arduino Nano and uses three sensors: an ultrasonic sensor to detect obstacles, a DHT11 sensor to measure temperature and humidity, and a water sensor to detect wet surfaces. The device gives alerts through a speaker, a buzzer, and a text-to-speech system. A Python Tkinter program shows the sensor data on a computer screen and provides voice feedback. The system also uses a CNN-based VGG19 model, trained on five common objects (almirah, chair, table, refrigerator, and TV), to recognize objects in real time through a webcam and announce them to the user. It was experimentally tested in different indoor environments under various lighting and noise conditions to estimate performance and reliability. The Smart Blind Stick is low-cost, modular, and easy to use. Its combination of sensors, AI, and human-computer interaction makes it a practical assistive tool.**

Keywords—Arduino Nano, HC-SR04 Ultrasonic Sensor, Water Sensor,DHT11 sensor, Obstacle Detection, Object Recognition, IoT, Visually Impaired

I. Introduction

According to the World Health Organization, there are 39 million blind people and 285 million visually impaired people in the world [3]. Everyday surroundings pose a challenge to this population, from evading obstacles to making out familiar objects or potential harm. Typical mobility devices, such as the white cane, are not that successful, relying solely on physical contact to identify obstacles and conveying no semantic nor contextual information regarding the surroundings. Visually impaired individuals resort to relying on other people's assistance or expensive proprietary systems that are often unavailable in developing countries. The latest technical breakthroughs in embedded systems, the Internet of Things (IoT), and artificial intelligence (AI) are opening up new prospects in intelligent assistive devices' design. These forms of technology enable the deployment of systems that are in a position to perceive, process, and act upon environmental stimulation in real-time. Among these assistive devices, smart blind sticks represent a new solution that attempts to expand individual autonomy for the visually challenged. This research explains the prototyping and design of a smart blind stick with microcontroller-based sensing, real-time processing, and deep learning-based object recognition with the goal to develop an inclusive assistive device. Ultrasonic sensing-based obstacle detection, weather sensing with the implementation of DHT11 sensors, water sensors for hazard detection, and camera-based object recognition through a CNN based model trained with common use daily household objects are integrated in the system. All the outputs of sensors are reported through a Python-based GUI with vocal output, giving an interactive experience designed in accordance with user needs. Researchers have done much research and many proposals to give an effective solution for visually impaired people so that they can overcome obstacles and be aware of danger. The proposed system, which we have implemented after reviewing some existing systems for the visually impaired [6]. We can say, as referred to in Wearable and Difficulty Finding Structure, which implemented two ultrasonic sensors, two IR sensors, a GSM, and a speaker. This system detects objects with ultrasonic and IR sensors, tracks object location with GSM, and gives feedback using a buzzer and voice alarm. Actually, they used the same sensor more than once. The aim of this project is to design, develop, and implement a smart blind stick that is more user-friendly, a multi-functional Smart Blind Stick system with embedded sensors and a deep learning- based vision model to assist visually impaired users in detecting obstacles, sensing conditions in the surroundings, and classifying objects in real-time, and developed with low manufacturing cost.

II. Literature Review

Many projects have been developed to assist visually impaired people, but most face limitations. Grover et al. [1] used three ultrasonic sensors, a soil moisture sensor, and Bluetooth with an Arduino UNO. The system detected obstacles well but was bulky and costly. Tirupal [2] in 2021 used four ultrasonic sensors and buzzers, which worked indoors but could not handle wet surfaces or recognize objects. Loganathan [3] in 2020 added infrared sensors for water detection, while Raghu [4] in 2024 included ultrasonic, IR, PIR, fire, and water sensors with vibration and voice alerts, but the design became complex. Srinivas [5] in 2025 added GPS, Wi-Fi, and dual Arduino boards, but again portability was an issue. Other works, such as Bele & Shubham [6] and Elsonbaty & Amira [7], also relied mainly on ultrasonic and water sensors with buzzer or GSM feedback.In computer vision, Masal & Mahadeo [8] used a VGG16 model for indoor objects with 92.53% accuracy, but performance dropped in noisy scenes. Stefan & Matcovici [9] trained an R-NN model on the COCO dataset with 95% accuracy, though limited data reduced its reliability. Zhang & Shaowei [10] proposed an LSTM-FCN model for corridor detection, but small objects were often missed due to dataset constraints. Overall, most systems rely heavily on ultrasonic and water sensors, with buzzer feedback. Many are complex, expensive, or lack voice guidance. Few use Arduino Nano, which is smaller and more efficient, and most do not provide real-time object recognition.

Our device takes a simpler approach. It uses an Arduino Nano with three sensors: an HC-SR04 ultrasonic sensor, a water sensor, and a DHT11. A buzzer and speaker provide both beeps and clear voice alerts about obstacles, wet surfaces, or extreme temperatures. Unlike others, it also recognizes

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objects in real time using a VGG19 model trained on a custom dataset and shows readings on a desktop app. The system is lightweight, affordable, and built for daily use.

Our Contribution

- Integrated three sensors (ultrasonic, water, and DHT11) with Arduino Nano.
- Developed a simple desktop app to display real-time outputs.
- Added voice feedback for natural alerts.
- Trained a custom VGG19 model for real-time object recognition.

III. Proposed System

The proposed system is portioned into three different portion. These are the hardware section, Object recognition and IoT integration.

A. Working Procedure

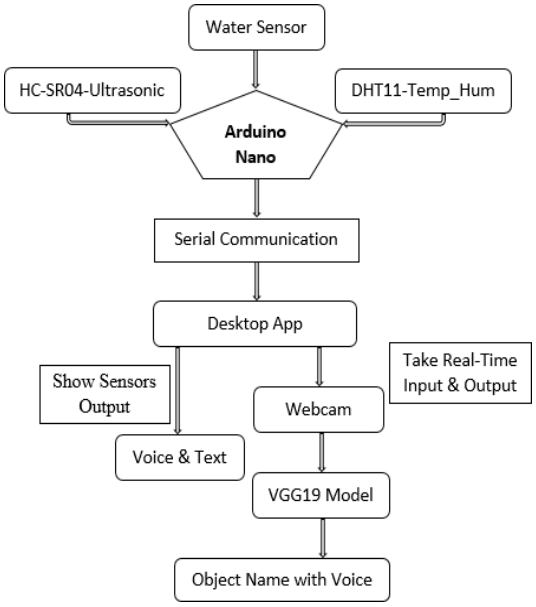


Fig 1: Block Diagram of System

HC-SR04, DHT11 and Water sensor are consequently check the surroundings and sends signal to the Arduino Nano. Webcam captures the real-time object and recognized the object using trained VGG19 model. Then Arduino combined two feedback and provide output text with voice in speaker.

a. Hardware Section

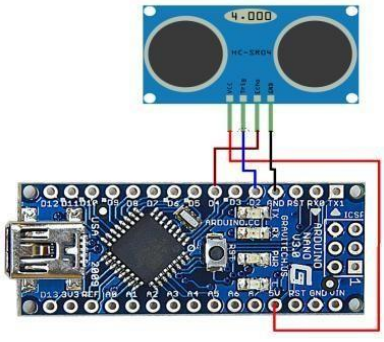


Fig 2 : HC-SR04 sensor integrate with Arduino Nano

In Fig 2 circuit diagram shows how to connect HC-SR04 ultrasonic with Arduino Nano and what task it will be performed. Ultrasonic sensor's VCC pin connect with 5V, Trig pin connect with D2, Echo pin connect with D3 and GND to GND of the Arduino Nano. These sensor provides signal to the Arduino if there is any object between 400 cm.

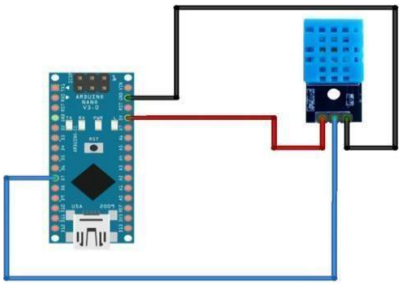


Fig 3 : DHT11 sensor integrate With Arduino Nano

In Fig 3 shows how to connect DHT11 with Arduino Nano. DHT11 VCC pin connect with 5V, Data pin connect with D4 and GND to GND of the Arduino Nano. This sensor mainly measures the surroundings real-time temperature and humidity which provides consciousness of the condition in environment for visually impaired people.

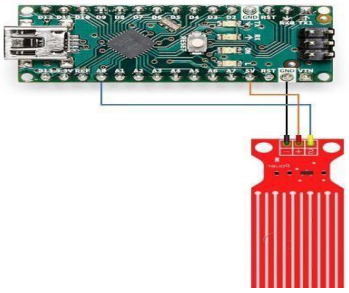


Fig 4 : Water sensor integrate Arduino Nano

In Fig 4 shows how to connect water sensor with Arduino Nano. Water sensor Signal(S) pin connect with A0, positive (+) pin connect with VCC, negative (-) connect with GND of the Arduino Nano. This sensor detect the presence of water, wet surfaces and water level and provides the information to the microcontroller which is very important for the blind people to move from one place to another safely.

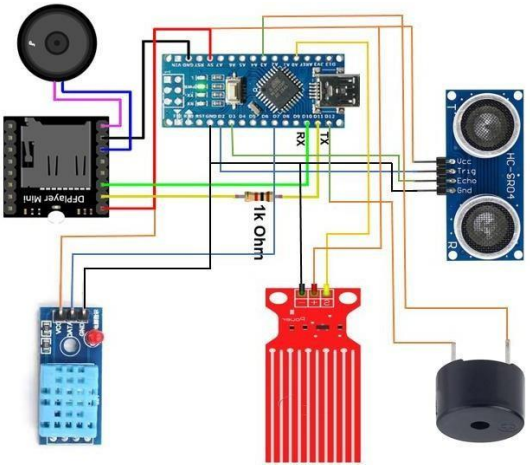


Fig 5 : Combined Circuit diagram of the system

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b. OBJECT RECOGNITION

A. Dataset

Our dataset has 18300 images of daily-life object. It includes 3000 images of chair, table, almirah, fridge and TV. Images are collected from different sources like phone picture and Google to ensure diversity and augmentation techniques (rotation, scaling, and brightness adjustment).We take photos of different sources in different folder. The data employed within this system is split into three sets: training, validation, and test. A total of 15000 images are used for training the model, 3000 images are used for validation and 300 were reserved for testing. Figure 6 is showing some images of our dataset.



Fig 6 : Dataset

It comprises five object classes: almirah, chair, fridge, table, and TV. Images are normalized to 224×224 pixels to cater to the input requirement of the VGG19 model and pixel value is normalized in the interval 0–1 to aid training performance. The training set is expanded to introduce variability, and performance of the model is assessed with the aid of the validation and test sets. This setup of the dataset allows for robust training and effective evaluation.

B. Pre-processing

These images are then run through preprocessing prior to training the model because all pictures were not uniform in size or format. To achieve uniformity as well as increased accuracy, all pictures were resized as well as all pictures given the same format. Figure-7 shows a very low-complexity picture is converted into an array representation.

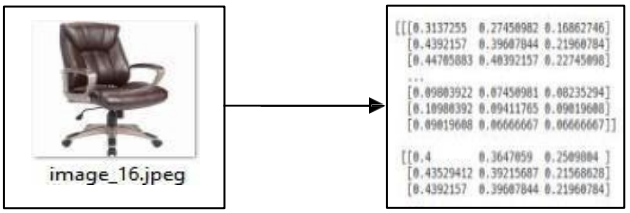


Fig 7 : Pre-processing

C. Apply VGG19 model

Here, a pre-trained convolutional neural network (CNN) model of VGG19 is utilized to identify surrounding objects such as almirah, chair, fridge, table, and TV. It is fine-tuned by self-defined dense and dropout layers, and the Adam optimizer is used along with TensorFlow and Keras to learn in an efficient manner. It consists of an architecture of convolution layers, pooling layers, and fully connected layers from the model of VGG19 to aid efficient feature extraction and classification. Overall 97% accuracy was obtained for the proposed model from the validation dataset, which is highly efficient to identify objects in real time for smart blind stick implementations. Figure 8 illustrates the architecture of the proposed VGG19-based CNN model.

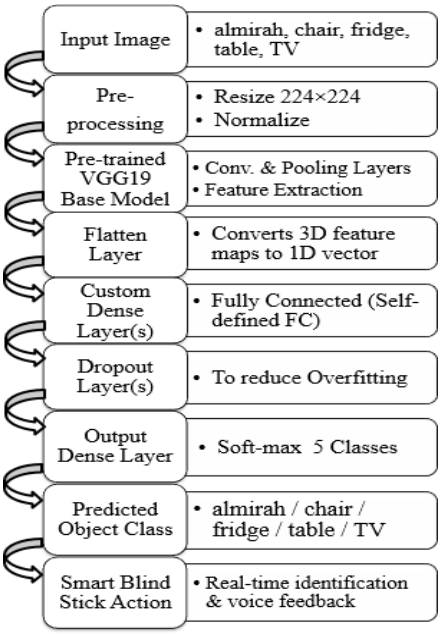


Fig 8 *:* Architecture of VGG19 model

D. Save the model

We can save the model in h5 format and later using this model, the daily life object detection system will be able to recognize the daily-life indoor and outdoor object

C. IOT INTEGRATION

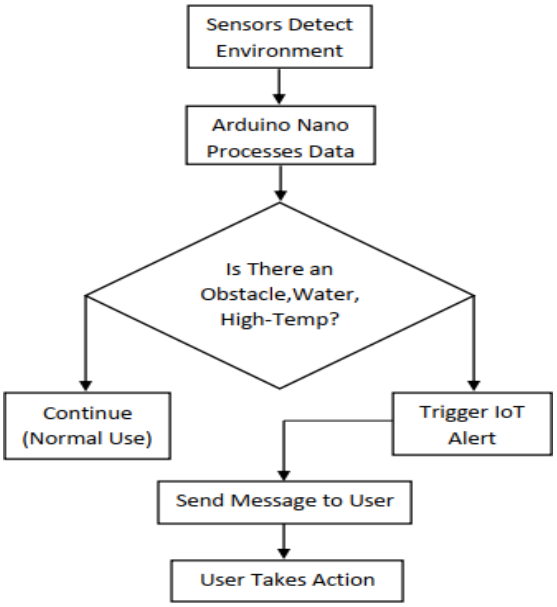


Fig 9: Diagram of IoT integration

With the help of IoT, the intelligent blind stick is able to keep track of the environment continuously and report status in a real-time basis. It adds automation, enhances safety, and

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provides reliability to the blind users. With the help of IoT, the stick is capable of sending a warning signal to the caretaker or family person in case of detecting an obstacle, hazard, or emergency situation. It makes the device more intelligent, prompt and convenient to use. The flow diagram in Figure 9 shows that the intelligent blind stick reports a message to the caretaker in case of detecting a hazard so that they might take immediate action.

IV. Result and Discussion

In this Project we built "The Smart Blind Stick" and kept it simple and clean. Then we tested it first in the lab, then in real life for both indoor and outer side of the house, wet floors also for tight corridors. The ultrasonic sensor spotted objects pretty well. Anything in the range of 1-400 cm got detected. The stick could say if object is very close (0–100 cm), Object is just near (101–200 cm), Object is present (201–400 cm). Each one had a different alert not only just a beep but also provides a real voice. People could tell what's coming. The water sensor was sharp too. We gave it a threshold value-500. Above that it says, "Wet floor" and buzzes with a special pattern. Also DHT11 sensor measures real-time temperature and humidity. If temperature is between 20-30 degrees this system could say temperature is normal, greater than 30 degree temperature is high and less than 20 degree temperature is cool. If humidity is 50-80% this stick could say normal humidity, for greater than 80% could say humidity is high, for less than 50% could say humidity is low and air is dry. In this way this system can successfully detect object, water, wet surfaces, recognize object and predict the real-time condition of air and weather of surroundings.

A. Experimental Setup and Testing Conditions

This Smart Blind Stick was experimentally tested in Lab and real-time indoor environments. Total 30 test were guided under various lighting and noise conditions to justify reliability. This system was conducted 15 indoor like room and corridor. Rest half was conducted in outdoor like as veranda and shaded area which was performed in different lighting conditions (Low-Light, normal room light and bright daylight) and noise levels (Quiet indoor and moderate background noise). This system was testing with three visually impaired people by walking with the stick in controlled paths. These users provide some important feedback which one was this system was suitable for outdoor object recognition that is already add the system update version. It is seemed adding this functionality increase the system performance, usability and reliability.

B. Sensor Limitations and Error Analysis

Several issues were made to affect system accuracy. HC-SR04 sensor showed small distance deviations when detecting reflective materials like metal, glass or soft surfaces that absorbed and weakened the echo signal. In ultrasonic processing strong ambient light also caused for slight delays. This ultrasonic sensor struggles with reflective surfaces due to sound waves bounce unpredictably, and soft materials absorb sound energy. Due to inconsistent electrode contact analog water sensor created occasional variations on uneven or partially wet surfaces. Analog water sensor shows variations due to uneven or slightly wet surfaces. The DHT11 sensor displayed its slow response time, resulting in 1-2 sec delay when temperature and humidity change rapidly. Its delay arises due to its low sampling rate and Precisions limitation. A short detection delay like as 0.2-0.3 sec made during moderate background noise.

In table-1 is showing different result evolution for our trained VGG19 model like as precision, recall and F1- Score are shown on this table.

Table 1 : Classification Report

Object Name	precision	recall	f1-score	support
almirah	0.97	0.98	0.98	6000
chair	0.98	0.98	0.98	6000
table	0.98	0.98	0.98	6000
fridge	0.98	0.96	0.97	6000
tv	0.97	0.97	0.97	6000
accuracy	**0.97**	**0.97**	**0.97**	**3000**

C. Confusion Matrix

In Fig 10 object recognition model performance is measured with a confusion matrix that tabulates the correspondence between the predicted and actual classes. The model predicted 587 images of almirah, 588 images of chair, 586 images of table, 579 images of fridge, and 583 images of tv from a total of 600 test images per class. From this comparison of predicted and actual labels, it is evident that the model is very accurate in all the object categories and it is easy to get a clear picture of where the model is strongest and where it might be weakest.

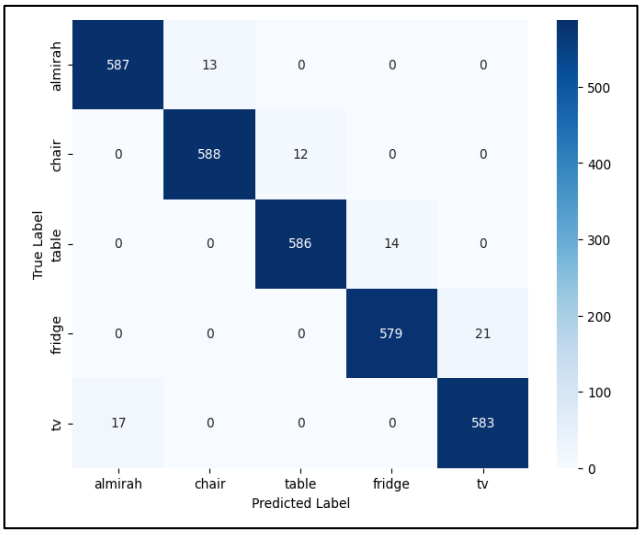


Fig 10 : Confusion Matrix

D. ROC Curve

In Fig 11 shows the ROC curve of the object recognition model. ROC curve is a visualization that illustrates the tradeoff between the True Positive Rate (TPR) and False Positive Rate (FPR) with different classification thresholds. Each of the curves is linked with a specific object category and reflects the ability of the model in distinguishing that category from the other classes. The more the curve is huddled in the top-left corner of the plot, the better the predictive capability of the model. From the visualization, it is easy to understand the ability of the model in terms of distinguishing objects in the right classes and evading false ones.

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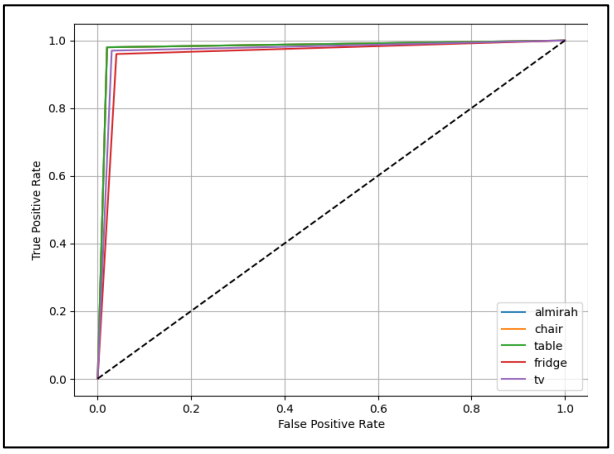


Fig 11 : ROC Curve

Table 2 : System Test Scenario and Result

Test ID	Scenario	Expected Outcome	Result
T01	Obstacle detected at 20 cm	Buzzer ON for 500 ms, Alarm voice ON	Success
T02	No obstacle in range	Buzzer remains OFF, Alarm voice OFF	Success
T03	Water detected (sensor > 300)	Buzzer beeps 3 times, Alarm voice ON	Success
T04	No water detected (sensor < 300)	Buzzer remains OFF, Alarm voice OFF	Success
T05	Temperature 25°C, Humidity 60%	Normal reading displayed, no alarm	Success
T06	Temperature > 40°C	Alarm triggered with buzzer ON (1000 ms)	Success
T07	Multiple hazards (obstacle + water)	Buzzer continuous beep + Alarm voice ON	Success

Table 3: Comparison of Existing Smart Blind Stick Systems with the Proposed Solution

System Name	Obstacle Recognition	Additional Features	Output System	Portability	Ease of use
[1] Smart Blind Stick	Detect Object ultrasonic not recognize object	Basic buzzer alert	Buzzer sound	Portable	Moderate(User needs training to interpret buzzer feedback)
[2] Smart Blind Stick using ultrasonic sensor	Can detect Obstacle with ultrasonic but not recognize object	Bluetooth module and buzzer for alert	Buzzer sound	Portable but not so helpful for users	Moderate(User needs training to interpret buzzer feedback)
[7] Smart Blind Stick Design and Implementation	For detecting obstacle use ultrasonic	Detect water and wet surfaces using water sensor	Buzzer sound	Portable	Require training to understand buzzer feedback
Proposed System(Our)	Can detect and recognize object by VGG19 model which is trained by own customize dataset	Water, detection, real-life temperature and humidity monitoring, AI-based object recognition model	Buzzer & Voice with object name	Highly portable for using local components	Very easy to use for it's simple design and this system gives us object distance with object name & buzzer will be beeped

Table 4 : Comparison with Existing System of Object

Research Work	Technique Used	Dataset Type	Accuracy	Limitations
[8] Masal et al. & K. Mahadeo (2024)	Used VGG16 for Feature Extraction	Indoor object	92.53%	Performance reduces for noisy scenes of limited contextual understanding
[9] Stefan & Matcovici (2023)	R-NN model	COCO Dataset	95%	Use Small amount of data(Don't detect object properly)
[10] Zhang & Shaowei (2023)	LSTM-FCN	Indoor dataset	96%	Focused long corridor object detection but not small
Our Proposed System (This Work)	VGG19 used for Feature Extraction	Own Customized Dataset	97%	Properly classify and recognized object

Table 2 presents seven test scenarios for the system with the expected output will be and corresponding feedback. Table 3 provides a comparative analysis between three different types of smart blind stick and our proposed system, highlighting advantages of our proposed system. Table 4 compares our object recognition model with three existing models. Which shows ours model performance.

Table 5 is providing an overview of our system like accuracy, hardware cost and more features.

Table 5 : System Summary

Feature	Result
VGG19 Accuracy	97%
Object Recognition	Recognize object
Hardware Cost	Comparatively lower cost
Local Component Usage	Parts are sourced locally

V. Conclusion

In conclusion, our Smart Blind Stick shows that affordable technology can make a real difference in the daily lives of visually impaired people. The Arduino Nano allows the system to run smoothly and continuously while keeping the device small and simple to carry. Real user feedback confirmed that voice alerts feel more natural and helpful than beeps or vibrations, which makes the device easier to use compared to many existing solutions.

There are still some challenges. A single ultrasonic sensor cannot always detect objects at an angle, and in noisy environments the audio warnings may be harder to hear. To improve this, we are working on adding GPS and GSM modules for outdoor navigation, as well as upgrading our VGG19-based model with a larger dataset that includes traffic signs, vehicles, pets, and other real-world obstacles.

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Even with its current limitations, the Smart Blind Stick has proven to be both practical and effective. Right now, this system is initially designed and developed for indoor environments and recognizes five classes of object(almirah, chair, table, fridge, and tv).For outdoor use need to add object classes like vehicles, pedestrians, and traffic signs will be integrated into update versions. More user which are visually impaired individuals testing will be conducted to estimate usability, comfort and real-time performance in daily life. This update version will increase the system's adaptability and ensure real-world deployment. With further development, it has strong potential to become a reliable everyday companion for visually impaired users.

REFERENCES

[1] S. Grover, A. Hassan, K. Yashaswi, and N. K. Shinde, "Smart Blind Stick," *Int. J. Electron. Commun. Eng.*, vol. 7, no. 5, pp. 19–23, May 2020, doi: 10.14445/23488549/IJECE-V7I5P104.

[2] T. Tirupal, B. V. Murali, M. Sandeep, K. S. Kumar, and C. U. Kumar, "Smart Blind Stick Using Ultrasonic Sensor," vol. 7, no. 2, 2021.

[3] N. Loganathan, K. Lakshmi, N. Chandrasekaran, S. R. Cibisakaravarthi, R. H. Priyanga, and K. H. Varthini, "Smart Stick for Blind People," in *2020 6th International Conference on Advanced Computing and Communication Systems (ICACCS)*, Coimbatore, India: IEEE, Mar. 2020, pp. 65–67. doi: 10.1109/ICACCS48705.2020.9074374.

[4] Raghu N., "Design and Implementation of Smart Blind Stick for Obstacle Detection and Navigation System," *J. Electr. Syst.*, vol. 20, no. 2s, pp. 1277–1289, Mar. 2024, doi: 10.52783/jes.1770.

[5] G. Srinivas, G. M. Raju, D. Ramesh, and S. Sivaram, "SMART BLIND STICK CONNECTED SYSTEM USING ARDUINO," vol. 6, no. 2.

[6] S. Bele, S. Ghule, A. Gunjal, and N. Anwat, "Design and Implementation of Smart Blind Stick," *SSRN Electron. J.*, 2020, doi: 10.2139/ssrn.3645413.

[7] Amira. A. Elsonbaty, "Smart Blind Stick Design and Implementation," *Int. J. Eng. Adv. Technol.*, vol. 10, no. 5, pp. 17–20, June 2021, doi: 10.35940/ijeat.D2535.0610521.

[8] K. M. Masal, S. Bhatlawande, and S. D. Shingade, "Attention-based Deep Learning model for Indoor Object Recognition Framework for Visually Impaired Individuals," in *2024 IEEE International Conference on Communication, Computing and Signal Processing (IICCCS)*, ASANSOL, India: IEEE, Sept. 2024, pp. 1–6. doi: 10.1109/IICCCS61609.2024.10763601.

[9] M. Stefan, D. Voinea, and A.-I. Popa, "$k-\text{NN}$ embeded space conditioning for enhanced few-shot object detection," in *2023 IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, Waikoloa, HI, USA: IEEE, Jan. 2023, pp. 401–410. doi: 10.1109/WACVW58289.2023.00044.

[10] S. Zhang *et al.*, "Long Short-Term Memory Fully Convolutional Network-Based Passing-Object Detection and Its Application to WLAN Sensing," in *2023 IEEE/CIC International Conference on Communications in China (ICCC Workshops)*, Dalian, China: IEEE, Aug.2023,pp.16.doi:10.1109/ICCCWorkshops57813.2023.10233733.

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